



Brief paper

Passivity-based synchronization of a class of complex dynamical networks with time-varying delay[☆]



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ABSTRACT

This paper proposes a complex delayed dynamical network consisting of N linearly and diffusively coupled identical reaction–diffusion neural networks. By utilizing some inequality techniques, a sufficient condition ensuring the output strict passivity is derived for the proposed network model. Then, we reveal the relationship between output strict passivity and synchronization of the proposed network model. Moreover, based on the obtained passivity result and the relationship between output strict passivity and synchronization, a criterion for synchronization is established. Finally, a numerical example is provided to illustrate the correctness and effectiveness of the proposed results.

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1. Introduction

Today various complex networks can be seen everywhere and are becoming an important part of our daily life. Some of the most well-known examples include food webs, communication networks, social networks, power grids, cellular networks, World Wide Web, metabolic systems, disease transmission networks, etc. Therefore, the topology and dynamical behavior of complex dynamical networks have been extensively studied by the researchers. In particular, the synchronization problem of complex dynamical networks has received much of the focus in recent years. So far, a great many important results on synchronization have been obtained for various complex networks (Arcak, 2011; Lü & Chen, 2005; Scardovi, Arcak, & Sontag, 2010; Slotine & Wang, 2005;

Yao, Guan, & Hill, 2009; Yao, Wang, Guan, & Xu, 2009; Zhou, Lu, & Lü, 2006, 2008). In Slotine and Wang (2005), the authors studied the synchronization of a network with fixed and switching topologies by using partial contraction theory. In Arcak (2011), a sufficient condition was obtained to guarantee the synchronization of a compartmental ODE model by utilizing the properties of the Laplacian matrix and the Mean-Value theorem. However, in these existing works, the node state is only dependent on the time. But, in many circumstances, the node state is not only dependent on the time, but also intensively dependent on space variable.

As a special class of complex networks, arrays of coupled neural networks have attracted much attention in recent years. Especially, the synchronization problem of arrays of coupled neural networks has stirred much research interest due to its fruitful applications in various fields. In Hoppensteadt and Izhikevich (2000), the authors proposed an architecture of coupled neural networks to store and retrieve complex oscillatory patterns as synchronization states. In Zhang and He (1997), a secure communication system based on coupled cellular neural networks was presented. In addition, the study of synchronization of coupled neural networks is an important step for understanding brain science (Gray, 1994; Ukhtomsky, 1978). Therefore, it is interesting to investigate the synchronization of coupled neural networks. In Tang and Fang (2009), the authors considered a general model of an array of N linearly coupled delayed neural networks with

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Markovian jumping hybrid coupling, which is composed of constant coupling, discrete and distributed time-varying delay coupling. In Li, Song, and Fei (2010), an array of coupled discrete-time Cohen–Grossberg neural networks with time-varying delay was discussed. However, in these existing works (Li et al., 2010; Tang & Fang, 2009), the diffusion effects have not been considered. Strictly speaking, diffusion effects cannot be avoided in neural networks when electrons are moving in asymmetric electromagnetic fields, thus we must consider the diffusion effects in neural networks. To our knowledge, very few researchers have investigated the synchronization in coupled reaction–diffusion neural networks (Liu, 2010; Wang, Teng, & Jiang, 2012). In Liu (2010), the author investigated a class of linearly coupled reaction–diffusion neural networks with unbounded time delays. In Wang, Teng et al. (2012), the adaptive synchronization in an array of linearly coupled neural networks with reaction–diffusion terms and time delays was studied.

Recently, passivity theory has also received a great deal of attention, and many results on this topic have been reported. The passivity theory was firstly proposed in the circuit analysis (Bevelevich, 1968), and since then has found successful applications in diverse areas such as stability, complexity, signal processing, chaos control and synchronization, fuzzy control, group coordination, power control, flow control, energy management, and so on (Arcak, 2007; Hill & Moylan, 1977; Wu, 2001; Xie, Fu, & Li, 1998). Although research on passivity has attracted so much attention, little of that had been devoted to the passivity properties of the spatially and temporally complex dynamical networks until Wang, Wu and Guo (Wang, Wu, & Guo, 2011) obtained the conditions for passivity of reaction–diffusion neural networks. To the best of our knowledge, the passivity of arrays of coupled reaction–diffusion neural networks has not yet been considered. Therefore, it is important and interesting to study the passivity of coupled reaction–diffusion neural networks. On the other hand, the passivity theory has long been a nice tool for analyzing the synchronization of the complex networks. But in most existing works, it is assumed that the node state is only dependent on the time. Therefore, it is essential to investigate the relationship between passivity and synchronization of the coupled reaction–diffusion neural networks.

The objective of this paper is to study the synchronization problem of arrays of coupled reaction–diffusion neural networks by using the passivity theory. The main contributions of this paper are as follows. First, we establish a criterion for the output strict passivity by utilizing some inequality techniques. Second, we reveal the relationship between output strict passivity and synchronization of the proposed network model. Third, by employing the obtained passivity result and the relationship between output strict passivity and synchronization, a sufficient condition for synchronization of the complex dynamical network is derived.

The rest of this paper is organized as follows. In Section 2, our mathematical model of complex network is presented and some preliminaries are given. The main results of this paper are given in Section 3. In Section 4, a numerical example is provided to illustrate the effectiveness of the theoretical results. Finally, Section 5 concludes the investigation.

2. Network model and preliminaries

Let $\mathbb{R} = (-\infty, +\infty)$, $\mathbb{R}^+ = [0, +\infty)$, $\mathbb{R}^n$ be the n -dimensional Euclidean space and $\mathbb{R}^{n \times m}$ be the space of $n \times m$ real matrices. $P \in \mathbb{R}^{n \times n} \geq 0$ ($P \in \mathbb{R}^{n \times n} \leq 0$) means that matrix P is symmetric and semi-positive (semi-negative) definite. $P \in \mathbb{R}^{n \times n} > 0$ ($P \in \mathbb{R}^{n \times n} < 0$) means that matrix P is symmetric and positive (negative) definite. I_n denotes the $n \times n$ real identity matrix. B^T denotes the transpose of matrix B . $\otimes$ denotes the Kronecker product of two matrices. $\lambda_m(\cdot)$ and $\lambda_M(\cdot)$ denote the minimum and the maximum eigenvalue of the corresponding matrix, respectively. $\Omega = \{x =$

$(x_1, x_2, \dots, x_q)^T \mid |x_k| < l_k, k = 1, 2, \dots, q\}$ is an open bounded domain in $\mathbb{R}^q$ with smooth boundary $\partial\Omega$, $\overline{\Omega} = \Omega \cup \partial\Omega$, and $\text{mes}\Omega$ denotes the measure of Ω . For any $e(x, t) = (e_1(x, t), e_2(x, t), \dots, e_n(x, t))^T \in \mathbb{R}^n$, $(x, t) \in \Omega \times \mathbb{R}$, $\|e(\cdot, t)\|_2$ denotes

$$\|e(\cdot, t)\|_2 = \left(\int_{\Omega} \sum_{i=1}^n e_i^2(x, t) dx \right)^{\frac{1}{2}}.$$

In addition, we define $\|e(\cdot, t)\|_{\tau} = \sup_{-\tau \leq \theta \leq 0} \|e(\cdot, t + \theta)\|_2$.

In this paper, we consider a complex dynamical network consisting of N identical reaction–diffusion neural networks. To facilitate the readers, the complex network model is presented in a step-by-step format.

A single reaction–diffusion neural network with Dirichlet boundary conditions is described by the following partial differential equations (PDEs):

$$\begin{aligned} \frac{\partial w_i(x, t)}{\partial t} &= d_i \Delta w_i(x, t) - a_i w_i(x, t) + J_i \\ &+ \sum_{j=1}^n b_{ij} f_j(w_j(x, t)) \end{aligned} \quad (1)$$

where $i = 1, 2, \dots, n$, n is the number of neurons in the network; $x = (x_1, x_2, \dots, x_q)^T \in \Omega \subset \mathbb{R}^q$; $w_i(x, t) \in \mathbb{R}$ is the state of the i th neuron at time t and in space x ; $\Delta = \sum_{k=1}^q \frac{\partial^2}{\partial x_k^2}$ is the Laplace diffusion operator on Ω ; $d_i > 0$ represents the transmission diffusion coefficient along the i th neuron; $f_j(\cdot)$ denotes the activation function of the j th neuron; $a_i > 0$ represents the rate with which the i th neuron will reset its potential to the resting state when disconnected from the network and external input; b_{ij} denotes the strength of the j th neuron on the i th neuron; J_i is a constant external input.

The initial value and boundary value conditions associated with system (1) are given in the form

$$w_i(x, 0) = \phi_i(x), \quad x \in \Omega, \quad (2)$$

$$w_i(x, t) = 0, \quad (x, t) \in \partial\Omega \times [0, +\infty) \quad (3)$$

where $\phi_i(x)$ ($i = 1, 2, \dots, n$) is bounded and continuous on Ω .

Throughout this paper, the function $f_j(\cdot)$ ($j = 1, 2, \dots, n$) satisfies the Lipschitz condition, that is, there exists positive constant ρ_j such that

$$|f_j(\xi_1) - f_j(\xi_2)| \leq \rho_j |\xi_1 - \xi_2|$$

for any $\xi_1, \xi_2 \in \mathbb{R}$, where $|\cdot|$ is the Euclidean norm.

We can rewrite system (1) in a compact form as follows:

$$\frac{\partial w(x, t)}{\partial t} = D \Delta w(x, t) - A w(x, t) + J + B f(w(x, t)) \quad (4)$$

where $D = \text{diag}(d_1, d_2, \dots, d_n)$, $B = (b_{ij})_{n \times n}$, $J = (J_1, J_2, \dots, J_n)^T$, $A = \text{diag}(a_1, a_2, \dots, a_n)$, $f(w(x, t)) = (f_1(w_1(x, t)), f_2(w_2(x, t)), \dots, f_n(w_n(x, t)))^T$, $w(x, t) = (w_1(x, t), w_2(x, t), \dots, w_n(x, t))^T$.

N mutually coupled reaction–diffusion neural networks (4) can result in a complex network, which is described by

$$\begin{aligned} \frac{\partial z_i(x, t)}{\partial t} &= D \Delta z_i(x, t) - A z_i(x, t) + B f(z_i(x, t)) \\ &+ c_1 \sum_{j=1}^N G_{ij}^1 \Gamma_1 z_j(x, t) + J \\ &+ c_2 \sum_{j=1}^N G_{ij}^2 \Gamma_2 z_j(x, t - \tau(t)) + u_i(x, t) \end{aligned} \quad (5)$$

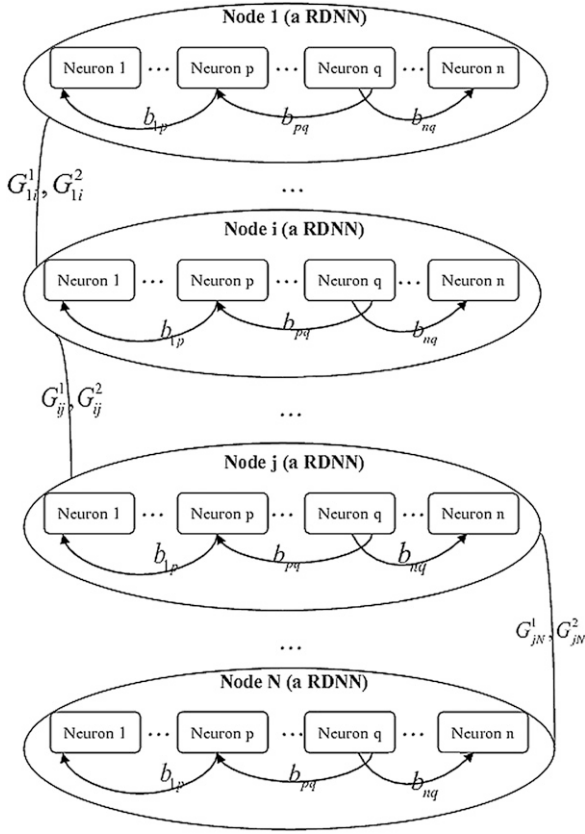


Fig. 1. Complex dynamical network (5) ($i, j \in \{1, 2, \dots, N\}$, $p, q \in \{1, 2, \dots, n\}$).

where $i = 1, 2, \dots, N$, N is the number of nodes in the network; $\tau(t)$ is the time-varying delay with $0 \leq \tau(t) \leq \tau$; $z_i(x, t) = (z_{i1}(x, t), z_{i2}(x, t), \dots, z_{in}(x, t))^T \in \mathbb{R}^n$ is the state vector of node i ; $u_i(x, t) \in \mathbb{R}^n$ is the control input; c_1 and c_2 are positive real numbers, which represent the coupling strength for non-delayed configuration and delayed one, respectively; $\Gamma_1 = \text{diag}(\gamma_1^1, \gamma_2^1, \dots, \gamma_n^1) \in \mathbb{R}^{n \times n}$ and $\Gamma_2 = \text{diag}(\gamma_1^2, \gamma_2^2, \dots, \gamma_n^2) \in \mathbb{R}^{n \times n}$ are positive definite matrices, which describe the individual coupling between two nodes for non-delayed configuration and delayed one, respectively; $G^1 = (G_{ij}^1)_{N \times N}$ and $G^2 = (G_{ij}^2)_{N \times N}$ represent the topological structure of network and coupling strength between nodes for non-delayed configuration and delayed one, respectively, where G_{ij}^1 (G_{ij}^2 can be defined similarly) is defined as follows: if there exists a connection between node i and node j , then $G_{ij}^1 = G_{ji}^1 > 0$; otherwise, $G_{ij}^1 = G_{ji}^1 = 0$ ($i \neq j$), and the diagonal elements of matrix G^1 are defined by

$$G_{ii}^1 = - \sum_{\substack{j=1 \\ j \neq i}}^N G_{ij}^1, \quad i = 1, 2, \dots, N.$$

In this paper, the topological structure of complex network (5) is undirected and weighted. The initial value and boundary value conditions associated with network (5) are given in the form

$$z_i(x, t) = \Phi_i(x, t) \in \mathbb{R}^n, \quad (x, t) \in \Omega \times [-\tau, 0], \quad (6)$$

$$z_i(x, t) = 0, \quad (x, t) \in \partial\Omega \times [-\tau, +\infty) \quad (7)$$

where $\Phi_i(x, t)$ ($i = 1, 2, \dots, N$) is a bounded and continuous function on $\Omega \times [-\tau, 0]$.

Remark 1. Many systems in nature and society can be modeled as complex networks, which consist of nodes connected by edges. In

a complex network, each node represents a fundamental unit with specific activity, while edges represent the relationship of these fundamental units. In this paper, we consider a complex dynamical network consisting of N identical nodes, in which each node is an n -dimensional reaction-diffusion neural network (RDNN) (see Fig. 1). In complex network (5), G_{ij}^1 and G_{ij}^2 denote the impact strength of the j th node on the i th node for non-delayed case and delayed case, respectively.

In what follows, we introduce some useful definitions.

Definition 2.1 (See Willems, 1972). A system with supply rate ϑ is said to be dissipative if there exists a nonnegative function $S : \mathbb{R}^+ \rightarrow \mathbb{R}^+$, called the storage function, such that

$$\int_{t_0}^{t_p} \vartheta(u, y) dt \geq S(t_p) - S(t_0)$$

for any $t_p, t_0 \in \mathbb{R}^+$ and $t_p \geq t_0$, where $u(x, t) \in \mathbb{R}^n$ and $y(x, t) \in \mathbb{R}^n$ are the input and output of the system at time t and in space x , respectively, $(x, t) \in \Omega \times \mathbb{R}^+$.

Definition 2.2 (See Hill & Moylan, 1976; Wang et al., 2011). A system is said to be strictly passive if it is dissipative with respect to

$$\begin{aligned} \vartheta(u, y) = & \int_{\Omega} y^T(x, t) u(x, t) dx - \gamma_1 \int_{\Omega} u^T(x, t) u(x, t) dx \\ & - \gamma_2 \int_{\Omega} y^T(x, t) y(x, t) dx \end{aligned}$$

for $\gamma_1 \geq 0, \gamma_2 \geq 0, \gamma_1 + \gamma_2 > 0$, where $u(x, t) \in \mathbb{R}^n$ and $y(x, t) \in \mathbb{R}^n$ are the input and output of the system at time t and in space x , respectively, $(x, t) \in \Omega \times \mathbb{R}^+$.

The system is said to be input-strictly passive if $\gamma_1 > 0$ and output-strictly passive if $\gamma_2 > 0$.

3. Passivity-based synchronization of the complex delayed dynamical network

Suppose that $w^*(x) = (w_1^*(x), w_2^*(x), \dots, w_n^*(x))^T$ is an equilibrium solution of the reaction-diffusion neural network (1), then it satisfies (3) and

$$0 = D\Delta w^*(x) - Aw^*(x) + J + Bf(w^*(x)). \quad (8)$$

Remark 2. In this section, we assume that $w^*(x)$ is an equilibrium solution of the reaction-diffusion neural network (1), but we do not require that the equilibrium solution be unique. Obviously, the uniqueness of equilibrium solution is also an important and interesting problem. Practically, in Lu (2007), a class of interval reaction-diffusion Hopfield neural networks with Dirichlet boundary conditions was considered, and reaction-diffusion neural network (1) is a special case of this model. The author obtained a sufficient condition for the uniqueness of the equilibrium solution in Lu (2007).

Next, we give the definition of synchronization for the complex dynamical network (5).

Definition 3.1. The complex network (5) is said to achieve synchronization if

$$\lim_{t \rightarrow +\infty} \|z_i(\cdot, t) - w^*(\cdot)\|_2 = 0 \quad \text{for all } i = 1, 2, \dots, N$$

under the condition that $u_i(x, t) = 0, i = 1, 2, \dots, N$.

Defining $e_i(x, t) = z_i(x, t) - w^*(x)$, then the dynamics of the error vector $e_i(x, t)$ is governed by the following equation:

$$\begin{aligned} \frac{\partial e_i(x, t)}{\partial t} = & D\Delta e_i(x, t) - Ae_i(x, t) + Bf(z_i(x, t)) \\ & - Bf(w^*(x)) + c_1 \sum_{j=1}^N G_{ij}^1 \Gamma_1 e_j(x, t) \\ & + c_2 \sum_{j=1}^N G_{ij}^2 \Gamma_2 e_j(x, t - \tau(t)) + u_i(x, t) \end{aligned} \quad (9)$$

where $i = 1, 2, \dots, N$.

In what follows, we establish the passivity of system (9). The output vector $y_i(x, t)$ of system (9) is defined as

$$y_i(x, t) = Fe_i(x, t) + Hu_i(x, t) \quad (10)$$

for the passivity scheme, where $F, H \in \mathbb{R}^{n \times n}$ are known real matrices, $i = 1, 2, \dots, N$.

For the convenience, we denote

$$\begin{aligned} \hat{D} &= \text{diag}(D, D, \dots, D), & \hat{A} &= \text{diag}(A, A, \dots, A), \\ \hat{B} &= \text{diag}(B, B, \dots, B), & \hat{F} &= \text{diag}(F, F, \dots, F), \\ \hat{H} &= \text{diag}(H, H, \dots, H), & \Theta &= \text{diag}(\rho_1^2, \rho_2^2, \dots, \rho_n^2). \end{aligned}$$

By using Kronecker product, we can rewrite systems (9) and (10) in a compact form as follows:

$$\begin{aligned} \frac{\partial e(x, t)}{\partial t} = & \hat{D}\Delta e(x, t) - \hat{A}e(x, t) + \hat{B}\hat{f}(z(x, t)) \\ & - \hat{B}\hat{f}(\hat{w}^*(x)) + c_1(G^1 \otimes \Gamma_1)e(x, t) \\ & + c_2(G^2 \otimes \Gamma_2)e(x, t - \tau(t)) + u(x, t) \\ y(x, t) = & \hat{F}e(x, t) + \hat{H}u(x, t) \end{aligned} \quad (11)$$

where $e(x, t) = (e_1^T(x, t), e_2^T(x, t), \dots, e_N^T(x, t))^T$, $u(x, t) = (u_1^T(x, t), u_2^T(x, t), \dots, u_N^T(x, t))^T$, $y(x, t) = (y_1^T(x, t), y_2^T(x, t), \dots, y_N^T(x, t))^T$, $\hat{f}(z(x, t)) = (f^T(z_1(x, t)), f^T(z_2(x, t)), \dots, f^T(z_N(x, t)))^T$, $\hat{f}(\hat{w}^*(x)) = (f^T(w^*(x)), f^T(w^*(x)), \dots, f^T(w^*(x)))^T$.

Lemma 3.1. Let $\dot{\tau}(t) \leq \sigma < 1$. If there exist matrices $\hat{P} = \text{diag}(P^1, P^2, \dots, P^N) \in \mathbb{R}^{nN \times nN} > 0$, $Q \in \mathbb{R}^{nN \times nN} > 0$ and a scalar $\gamma > 0$ such that

$$\hat{P}\hat{D} + \hat{D}\hat{P} \geq 0 \quad (12)$$

$$\begin{pmatrix} \chi & \hat{P} - \hat{F}^T + \gamma\hat{F}^T\hat{H} \\ \hat{P} - \hat{F} + \gamma\hat{H}^T\hat{F} & -\hat{H} - \hat{H}^T + \gamma\hat{H}^T\hat{H} \end{pmatrix} \leq 0 \quad (13)$$

where $\hat{\Theta} = \text{diag}(\Theta, \Theta, \dots, \Theta)$, $\chi = -\sum_{k=1}^q \frac{1}{l_k^2} (\hat{P}\hat{D} + \hat{D}\hat{P}) - \hat{P}\hat{A} - \hat{A}\hat{P} + \hat{\Theta} + \hat{P}\hat{B}\hat{B}^T\hat{P} + c_1\hat{P}(G^1 \otimes \Gamma_1) + c_1(G^1 \otimes \Gamma_1)\hat{P} + \frac{Q}{1-\sigma} + c_2^2\hat{P}(G^2 \otimes \Gamma_2)Q^{-1}(G^2 \otimes \Gamma_2)\hat{P} + \gamma\hat{F}^T\hat{F}$, then system (11) is output-strictly passive in the sense of Definition 2.2.

Proof. See Appendix B.

Remark 3. To our knowledge, very few researchers have discussed the passivity of reaction-diffusion neural networks (Wang et al., 2011), in which the input and output variables are varied with the time and space variables. Especially, the passivity of coupled reaction-diffusion neural networks has not yet been investigated. In this paper, we consider a complex network consisting of N linearly and diffusively coupled identical reaction-diffusion neural networks, which is totally different from the network model considered in Wang et al. (2011). In Lemma 3.1, a sufficient condition ensuring output strict passivity is established, which can be applied to analyze linearly coupled reaction-diffusion neural networks (5) with output vector defined by (10).

Lemma 3.2. Suppose that $V : \mathbb{R}^+ \times \mathbb{C} \rightarrow \mathbb{R}^+$ is continuously differentiable and satisfies the following condition

$$v_1(\|e(\cdot, t)\|_2) \leq V(t, e_t(x, \theta)) \leq v_2(\|e(\cdot, t)\|_2) \quad (14)$$

where $e_t(x, \theta) \triangleq e(x, t + \theta)$, $-\tau \leq \theta \leq 0$, $v_1, v_2 : \mathbb{R}^+ \rightarrow \mathbb{R}^+$ are continuous and strictly monotonically nondecreasing functions, $v_1(s)$ and $v_2(s)$ are positive for $s > 0$ with $v_1(0) = v_2(0) = 0$. The complex network (5) is locally synchronized in the sense of Definition 3.1 if system (11) is output-strictly passive with respect to storage function V and matrix $\hat{F}$ is nonsingular.

Proof. System (11) is output-strictly passive with respect to storage function V , then there obviously exists a positive constant γ such that

$$\begin{aligned} V(t + \varepsilon) - V(t) \leq & \int_t^{t+\varepsilon} \int_{\Omega} y^T(x, s)u(x, s)dxds \\ & - \gamma \int_t^{t+\varepsilon} \int_{\Omega} y^T(x, s)y(x, s)dxds \end{aligned} \quad (15)$$

for any $t \in \mathbb{R}^+$ and $\varepsilon > 0$. Obviously, we can derive from (15) that

$$\begin{aligned} \frac{V(t + \varepsilon) - V(t)}{\varepsilon} \leq & \frac{\int_t^{t+\varepsilon} \int_{\Omega} y^T(x, s)u(x, s)dxds}{\varepsilon} \\ & - \gamma \frac{\int_t^{t+\varepsilon} \int_{\Omega} y^T(x, s)y(x, s)dxds}{\varepsilon}. \end{aligned} \quad (16)$$

By taking limit $\varepsilon \rightarrow 0$ in (16), we have

$$\dot{V}(t) \leq \int_{\Omega} y^T(x, t)u(x, t)dx - \gamma \int_{\Omega} y^T(x, t)y(x, t)dx. \quad (17)$$

Letting $u(x, t) = 0$, we can get from (17) that

$$\dot{V}(t) \leq -\gamma\lambda_m(\hat{F}^T\hat{F})\|e(\cdot, t)\|_2^2.$$

From Lemma A.2, we can conclude that system (11) is internally locally asymptotically stable. Therefore, the complex network (5) is locally synchronized in the sense of Definition 3.1. The proof is completed.

From Lemmas 3.1 and 3.2, we can obtain the following conclusion.

Theorem 3.1. Let $\dot{\tau}(t) \leq \sigma < 1$. If there exist matrices $F, H \in \mathbb{R}^{n \times n}$, $\hat{P} = \text{diag}(P^1, P^2, \dots, P^N) \in \mathbb{R}^{nN \times nN} > 0$, $Q \in \mathbb{R}^{nN \times nN} > 0$ and a scalar $\gamma > 0$ such that

$$\hat{P}\hat{D} + \hat{D}\hat{P} \geq 0 \quad (18)$$

$$\begin{pmatrix} \chi & \hat{P} - \hat{F}^T + \gamma\hat{F}^T\hat{H} \\ \hat{P} - \hat{F} + \gamma\hat{H}^T\hat{F} & -\hat{H} - \hat{H}^T + \gamma\hat{H}^T\hat{H} \end{pmatrix} \leq 0 \quad (19)$$

where F is a nonsingular matrix, $\hat{\Theta} = \text{diag}(\Theta, \Theta, \dots, \Theta)$, $\hat{F} = \text{diag}(F, F, \dots, F)$, $\hat{H} = \text{diag}(H, H, \dots, H)$, $\chi = -\sum_{k=1}^q \frac{1}{l_k^2} (\hat{P}\hat{D} + \hat{D}\hat{P}) - \hat{P}\hat{A} - \hat{A}\hat{P} + \hat{\Theta} + \hat{P}\hat{B}\hat{B}^T\hat{P} + c_1\hat{P}(G^1 \otimes \Gamma_1) + c_1(G^1 \otimes \Gamma_1)\hat{P} + \frac{Q}{1-\sigma} + c_2^2\hat{P}(G^2 \otimes \Gamma_2)Q^{-1}(G^2 \otimes \Gamma_2)\hat{P} + \gamma\hat{F}^T\hat{F}$, then complex network (5) is locally synchronized in the sense of Definition 3.1.

Remark 4. In Chopra (2012) and Chopra and Spong (2006), the authors studied the output synchronization of multi-agent systems, in which the agents' dynamics are input-output passive and described by affine nonlinear ODE systems. In this paper, we propose a complex delayed dynamical network consisting of N identical reaction-diffusion neural networks. Firstly, we analyze the output strict passivity of the proposed network model, which cannot be dealt with by those techniques used in

Chopra (2012) and Chopra and Spong (2006). Then, we reveal the relationship between output strict passivity and synchronization of the complex delayed dynamical network (5). Moreover, by using the obtained passivity result and the relationship between output strict passivity and synchronization, a sufficient condition ensuring synchronization is obtained.

Remark 5. To our knowledge, there are very few works on the synchronization of coupled reaction–diffusion neural networks (Liu, 2010; Wang, Teng et al., 2012), and the asymptotic stability of synchronization error system has not yet been investigated. In Theorem 3.1, a sufficient condition is established to ensure that the synchronization error system is locally asymptotically stable.

4. Numerical example

In this section, an illustrative example is provided to verify the effectiveness of the proposed theoretical results.

Consider a complex network consisting of 5 identical nodes with diffusive coupling, in which each node is a 3-dimensional reaction–diffusion neural network described by

$$\begin{aligned} \frac{\partial w_i(x, t)}{\partial t} = & d_i \Delta w_i(x, t) - a_i w_i(x, t) + J_i \\ & + \sum_{j=1}^3 b_{ij} f_j(w_j(x, t)) \end{aligned} \quad (20)$$

where $i = 1, 2, 3$, $\Omega = \{x \mid -0.5 < x < 0.5\}$, $f_j(\xi) = \frac{|\xi+1|-|\xi-1|}{4}$, $d_1 = 0.3$, $d_2 = 0.2$, $d_3 = 0.4$, $a_1 = 0.4$, $a_2 = 0.3$, $a_3 = 0.3$, $J_1 = J_2 = J_3 = 0$, and the matrix $B = (b_{ij})_{3 \times 3}$ is chosen as

$$B = \begin{pmatrix} 0.2 & 0.2 & 0.3 \\ 0.1 & 0.3 & 0.3 \\ 0.3 & 0.1 & 0.2 \end{pmatrix}.$$

Obviously, $w^*(x) = (0, 0, 0)^T \in \mathbb{R}^3$ is an equilibrium solution of the reaction–diffusion neural network (20), and $f_j(\cdot)$ ($j = 1, 2, 3$) satisfies the Lipschitz condition with $\rho_j = \frac{1}{2}$. In the following, we analyze the asymptotic stability of synchronized state $[(w^*(x))^T, (w^*(x))^T, \dots, (w^*(x))^T]^T$.

We take $c_1 = 0.2$, $c_2 = 0.1$, $\Gamma_1 = \text{diag}(0.4, 0.2, 0.3)$, $\Gamma_2 = \text{diag}(0.3, 0.1, 0.2)$ and $\tau(t) = 0.1 - 0.1e^{-t}$. The matrices G^1 and G^2 are chosen as, respectively,

$$G^1 = \begin{pmatrix} -0.5 & 0.2 & 0 & 0.3 & 0 \\ 0.2 & -0.8 & 0.2 & 0.2 & 0.2 \\ 0 & 0.2 & -0.5 & 0 & 0.3 \\ 0.3 & 0.2 & 0 & -0.5 & 0 \\ 0 & 0.2 & 0.3 & 0 & -0.5 \end{pmatrix},$$

$$G^2 = \begin{pmatrix} -0.3 & 0.1 & 0 & 0.2 & 0 \\ 0.1 & -0.6 & 0.2 & 0.1 & 0.2 \\ 0 & 0.2 & -0.4 & 0 & 0.2 \\ 0.2 & 0.1 & 0 & -0.3 & 0 \\ 0 & 0.2 & 0.2 & 0 & -0.4 \end{pmatrix}.$$

By using the YALMIP Toolbox of MATLAB, we can find the following matrices:

$$F = \begin{pmatrix} 0.8605 & -0.0401 & -0.0185 \\ -0.0427 & 1.0158 & -0.0226 \\ -0.0178 & -0.0204 & 0.7640 \end{pmatrix},$$

$$H = \begin{pmatrix} 0.6442 & 0.0048 & 0.0023 \\ 0.0048 & 0.6406 & 0.0025 \\ 0.0023 & 0.0025 & 0.6471 \end{pmatrix},$$

$$P = I_5 \otimes \begin{pmatrix} 1.1401 & -0.0413 & -0.0153 \\ -0.0413 & 1.3780 & -0.0203 \\ -0.0153 & -0.0203 & 1.0000 \end{pmatrix} > 0,$$

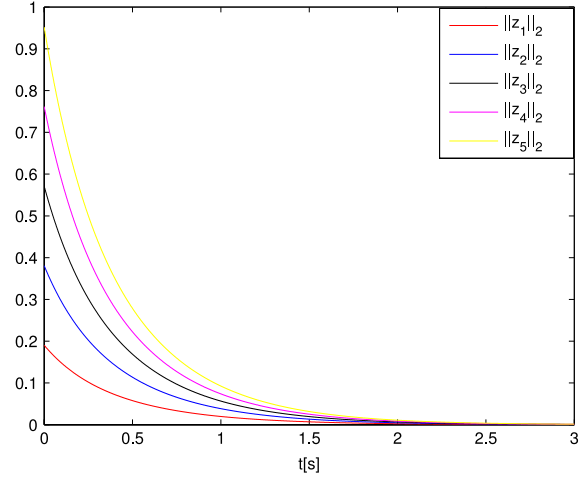


Fig. 2. The change processes of $\|z_i(\cdot, t)\|_2$, $i = 1, 2, \dots, 5$.

$$Q = I_5 \otimes \begin{pmatrix} 1.2851 & -0.1148 & -0.0773 \\ -0.1148 & 0.8580 & -0.0708 \\ -0.0773 & -0.0708 & 1.4187 \end{pmatrix} > 0$$

satisfying (18) and (19) with $\gamma = 0.4127$. According to Theorem 3.1, the complex network (5) with above given parameters is locally synchronized in the sense of Definition 3.1. The simulation results are shown in Fig. 2.

From Fig. 2, we clearly see that $\|z_i(\cdot, t)\|_2$, $i = 1, 2, \dots, 5$ are very close to 0 when the time t increases gradually to 2.5s, and this state is maintained along with the increasing of the time. These results show that $z_i(x, t)$ ($i = 1, 2, \dots, 5$) asymptotically converges to $w^*(x) = (0, 0, 0)^T$, thus the complex network (5) is synchronized. Fig. 2 demonstrates the correctness and effectiveness of the obtained result in Theorem 3.1.

Remark 6. It is obvious that the delay has a strong influence on the rate of change of the node state (see (5)). The change processes of $\|z_i(\cdot, t)\|_2$, $i = 1, 2, \dots, 5$, may be different if we take other time-varying delays in Fig. 2, although we omit this comparison for brevity. On the other hand, since $\sum_{j=1}^N G_{ij}^1 = 0$ and $\sum_{j=1}^N G_{ij}^2 = 0$ ($i = 1, 2, \dots, N$), $\sum_{j=1}^N G_{ij}^1 \Gamma_1 z_j(x, t)$ and $\sum_{j=1}^N G_{ij}^2 \Gamma_2 z_j(x, t - \tau(t))$ will tend to zero asymptotically if the complex network (5) is synchronized.

Remark 7. YALMIP Toolbox is a very convenient tool to solve linear matrix inequality (LMI), which has been widely used in analyzing the stability of various systems. Using the Schur complement, we can obtain that (19) is equivalent to

$$\begin{pmatrix} \mathbb{W}_1 + \frac{Q}{1-\sigma} & \hat{P} - \hat{F}^T & \hat{F}^T & \hat{P}\hat{B} & \mathbb{W}_2 \\ \hat{P} - \hat{F} & -\hat{H} - \hat{H}^T & \hat{H}^T & 0 & 0 \\ \hat{F} & \hat{H} & -\frac{1}{\gamma} I_{nN} & 0 & 0 \\ \hat{B}^T \hat{P} & 0 & 0 & -I_{nN} & 0 \\ \mathbb{W}_2^T & 0 & 0 & 0 & -Q \end{pmatrix} \leq 0,$$

where $\mathbb{W}_1 = -\sum_{k=1}^q \frac{1}{l_k^2} (\hat{P}\hat{D} + \hat{D}\hat{P}) - \hat{P}\hat{A} - \hat{A}\hat{P} + \hat{G} + c_1 \hat{P}(G^1 \otimes \Gamma_1) + c_1(G^1 \otimes \Gamma_1)\hat{P}$, $\mathbb{W}_2 = c_2 \hat{P}(G^2 \otimes \Gamma_2)$. Then, by utilizing the YALMIP Toolbox of MATLAB, we can find the $F, H, \hat{P}, Q, \gamma$ satisfying (18) and the above LMI.

5. Conclusion

In this paper, a general array model of coupled reaction–diffusion neural networks with delay coupling has been introduced. The output strict passivity of the proposed network model has been taken into consideration, and a sufficient condition has been established. Furthermore, the relationship between output strict passivity and synchronization of the proposed network model has been revealed. In addition, a sufficient condition ensuring synchronization has been derived by utilizing the obtained passivity result and the relationship between output strict passivity and synchronization. A numerical example has been provided to verify the correctness and effectiveness of the theoretical results. In future work, we shall study the passivity of coupled reaction–diffusion neural networks with impulsive effects.

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Appendix A. Some useful lemmas

Lemma A.1 (See [Lu, 2008](#)). Let Ω be a cube $|x_k| < l_k (k = 1, 2, \dots, q)$ and let $h(x)$ be a real-valued function belonging to $C^1(\Omega)$ which vanishes on the boundary $\partial\Omega$ of Ω , i.e., $h(x)|_{\partial\Omega} = 0$. Then

$$\int_{\Omega} h^2(x) dx \leq l_k^2 \int_{\Omega} \left(\frac{\partial h}{\partial x_k} \right)^2 dx$$

where $x = (x_1, x_2, \dots, x_q)^T$.

Consider the following delay partial differential equation (DPDE) system

$$\begin{cases} \frac{\partial \varphi(x, t)}{\partial t} = D\Delta\varphi(x, t) + g(\varphi(x, t - \tau(t))) + f(\varphi(x, t)) \\ \varphi(x, t) = \phi(x, t), \quad (x, t) \in \Omega \times [-\tau, 0] \\ \varphi(x, t) = 0, \quad (x, t) \in \partial\Omega \times [-\tau, +\infty) \end{cases} \quad (21)$$

where $x = (x_1, x_2, \dots, x_q)^T \in \Omega \subset \mathbb{R}^q$; $\varphi(x, t) = (\varphi_1(x, t), \varphi_2(x, t), \dots, \varphi_n(x, t))^T \in \mathbb{R}^n$ is the state vector of system at time t and in space x ; $\Delta = \sum_{k=1}^q \frac{\partial^2}{\partial x_k^2}$ is the Laplace operator on Ω ; $\tau(t)$ is the time-varying delay with $0 \leq \tau(t) \leq \tau$; $D = \text{diag}(d_1, d_2, \dots, d_n) > 0$; $\phi(x, t) \in \mathbb{R}^n$ is bounded and continuous on $\Omega \times [-\tau, 0]$; the functions $f(\cdot) \in \mathbb{R}^n$ and $g(\cdot) \in \mathbb{R}^n$ satisfy $f(0) = 0, g(0) = 0$, and the following assumption:

(A1) The functions $f(\cdot)$ and $g(\cdot)$ are continuous, and there exist two positive constants α_1 and α_2 such that

$$|f(\xi_1) - f(\xi_2)| \leq \alpha_1 |\xi_1 - \xi_2|,$$

$$|g(\xi_1) - g(\xi_2)| \leq \alpha_2 |\xi_1 - \xi_2|$$

for any $\xi_1, \xi_2 \in \mathbb{R}^n$, where $|\cdot|$ is the Euclidean norm.

Definition A.1. Let $\varphi(x, t, \phi)$ be the state trajectory of system (21) with initial condition $\phi(x, t)$. The equilibrium solution $\varphi^* = 0$ of system (21) is said to be asymptotically stable if for any $\epsilon > 0$, there exists $\delta(\epsilon)$ such that $\|\phi(\cdot, 0)\|_{\tau} < \delta(\epsilon)$ implies $\|\varphi(\cdot, t, \phi)\|_2 < \epsilon$ for $t \geq 0$, and there is a $b_0 > 0$ such that $\|\phi(\cdot, 0)\|_{\tau} < b_0$ implies $\|\varphi(\cdot, t, \phi)\|_2 \rightarrow 0$ as $t \rightarrow +\infty$.

To our knowledge, there are very few works on the asymptotic stability (in the sense of [Definition A.1](#)) of spatially and temporally complex dynamical networks. It is well known that Lyapunov

functional is a very convenient tool to analyze the asymptotic stability of DODE systems. In the following, we give a sufficient condition for the asymptotic stability of the equilibrium solution $\varphi^* = 0$ of system (21).

Lemma A.2 (See [Wang, Wu, & Guo, 2012](#)). Suppose that $v_1, v_2, v_3 : \mathbb{R}^+ \rightarrow \mathbb{R}^+$ are continuous and strictly monotonically nondecreasing functions, $v_1(s), v_2(s), v_3(s)$ are positive for $s > 0$ with $v_1(0) = v_2(0) = 0$. If there is a continuous functional $V : \mathbb{R}^+ \times C \rightarrow \mathbb{R}^+$ such that

$$v_1(\|\varphi(\cdot, t)\|_2) \leq V(t, \varphi_t(x, \theta)) \leq v_2(\|\varphi(\cdot, t)\|_{\tau}), \\ \dot{V}(t, \varphi_t(x, \theta)) \leq -v_3(\|\varphi(\cdot, t)\|_2)$$

where $\varphi_t(x, \theta) \triangleq \varphi(x, t + \theta)$, $-\tau \leq \theta \leq 0$, $\dot{V}$ is the derivative of V along the solution of system (21), then the equilibrium solution $\varphi^* = 0$ of system (21) is asymptotically stable.

Appendix B. Proof of Lemma 3.1

First, construct a Lyapunov functional for system (11) as follows:

$$V(t) = \int_{\Omega} e^T(x, t) \hat{P} e(x, t) dx \\ + \frac{1}{1 - \sigma} \int_{t-\tau(t)}^t \int_{\Omega} e^T(x, s) Q e(x, s) dx ds. \quad (22)$$

Calculating the time derivative of $V(t)$ along the trajectory of system (11), we can get

$$\begin{aligned} \dot{V}(t) &= 2 \int_{\Omega} e^T(x, t) \hat{P} \frac{\partial e(x, t)}{\partial t} dx + \frac{1}{1 - \sigma} \int_{\Omega} e^T(x, t) Q e(x, t) dx \\ &\quad - \frac{1 - \dot{\tau}(t)}{1 - \sigma} \int_{\Omega} e^T(x, t - \tau(t)) Q e(x, t - \tau(t)) dx \\ &\leq \int_{\Omega} [e^T(x, t) \hat{P} \hat{D} \Delta e(x, t) + (\Delta e(x, t))^T \hat{D} \hat{P} e(x, t) \\ &\quad + e^T(x, t) (-\hat{P} \hat{A} - \hat{A} \hat{P} + c_1 \hat{P} (G^1 \otimes \Gamma_1) + \frac{Q}{1 - \sigma} \\ &\quad + c_1 (G^1 \otimes \Gamma_1) \hat{P}) e(x, t) + 2e^T(x, t) \hat{P} \hat{B} (\hat{f}(z(x, t)) \\ &\quad - \hat{f}(\hat{w}^*(x))) + 2e^T(x, t) \hat{P} u(x, t) + 2c_2 e^T(x, t) \hat{P} \\ &\quad \times (G^2 \otimes \Gamma_2) e(x, t - \tau(t))] dx \\ &\quad - \int_{\Omega} e^T(x, t - \tau(t)) Q e(x, t - \tau(t)) dx. \end{aligned}$$

From Green's formula and the boundary condition, we have

$$\int_{\Omega} e_{il}(x, t) \Delta e_{ij}(x, t) dx = - \sum_{k=1}^q \int_{\Omega} \frac{\partial e_{il}(x, t)}{\partial x_k} \frac{\partial e_{ij}(x, t)}{\partial x_k} dx$$

where $l, j \in \{1, 2, \dots, n\}$, $i = 1, 2, \dots, N$. Letting $P^i = (p_{jl}^i)_{n \times n}$, we can obtain

$$\begin{aligned} &\int_{\Omega} e^T(x, t) \hat{P} \hat{D} \Delta e(x, t) dx \\ &= \sum_{i=1}^N \int_{\Omega} e_i^T(x, t) P^i D \Delta e_i(x, t) dx \\ &= \sum_{i=1}^N \sum_{j=1}^n \sum_{l=1}^n p_{jl}^i d_l \int_{\Omega} e_{ij}(x, t) \Delta e_{il}(x, t) dx \\ &= - \sum_{k=1}^q \sum_{i=1}^N \sum_{j=1}^n \sum_{l=1}^n p_{jl}^i d_l \int_{\Omega} \frac{\partial e_{ij}(x, t)}{\partial x_k} \frac{\partial e_{il}(x, t)}{\partial x_k} dx \end{aligned}$$

$$\begin{aligned}
&= - \sum_{k=1}^q \int_{\Omega} \left(\frac{\partial e(x, t)}{\partial x_k} \right)^T \hat{P} \hat{D} \frac{\partial e(x, t)}{\partial x_k} dx \\
&\quad \int_{\Omega} (\Delta e(x, t))^T \hat{D} \hat{P} e(x, t) dx \\
&= \sum_{i=1}^N \int_{\Omega} (\Delta e_i(x, t))^T D P^i e_i(x, t) dx \\
&= \sum_{i=1}^N \sum_{j=1}^n \sum_{l=1}^n d_{jl} p_{jl}^i \int_{\Omega} e_{il}(x, t) \Delta e_{ij}(x, t) dx \\
&= - \sum_{k=1}^q \int_{\Omega} \left(\frac{\partial e(x, t)}{\partial x_k} \right)^T \hat{D} \hat{P} \frac{\partial e(x, t)}{\partial x_k} dx.
\end{aligned}$$

Then, we can get

$$\begin{aligned}
&\int_{\Omega} (e^T(x, t) \hat{P} \hat{D} \Delta e(x, t) + (\Delta e(x, t))^T \hat{D} \hat{P} e(x, t)) dx \\
&= - \sum_{k=1}^q \int_{\Omega} \left(\frac{\partial e(x, t)}{\partial x_k} \right)^T (\hat{P} \hat{D} + \hat{D} \hat{P}) \frac{\partial e(x, t)}{\partial x_k} dx.
\end{aligned}$$

On the other hand, there obviously exists a real matrix $C \in \mathbb{R}^{nN \times nN}$ such that

$$\hat{P} \hat{D} + \hat{D} \hat{P} = C^T C,$$

then

$$\begin{aligned}
\left(\frac{\partial e(x, t)}{\partial x_k} \right)^T (\hat{P} \hat{D} + \hat{D} \hat{P}) \frac{\partial e(x, t)}{\partial x_k} &= \left(\frac{\partial e(x, t)}{\partial x_k} \right)^T C^T C \frac{\partial e(x, t)}{\partial x_k} \\
&= \left(\frac{\partial (C e(x, t))}{\partial x_k} \right)^T \frac{\partial (C e(x, t))}{\partial x_k}.
\end{aligned}$$

Let $\vartheta(x, t) = C e(x, t)$, for $(x, t) \in \partial \Omega \times [-\tau, +\infty)$ from the boundary condition (7), we have $\vartheta(x, t) = C e(x, t) = 0$. In view of Lemma A.1, one has

$$\begin{aligned}
&\sum_{k=1}^q \int_{\Omega} \left(\frac{\partial \vartheta(x, t)}{\partial x_k} \right)^T \frac{\partial \vartheta(x, t)}{\partial x_k} dx \\
&\geq \sum_{k=1}^q \frac{1}{l_k^2} \int_{\Omega} \vartheta^T(x, t) \vartheta(x, t) dx \\
&= \sum_{k=1}^q \frac{1}{l_k^2} \int_{\Omega} e^T(x, t) (\hat{P} \hat{D} + \hat{D} \hat{P}) e(x, t) dx. \tag{23}
\end{aligned}$$

Furthermore, we can easily derive

$$\begin{aligned}
&2e^T(x, t) \hat{P} \hat{B} \hat{f}(z(x, t)) - \hat{f}(\hat{w}^*(x)) \\
&= 2 \sum_{i=1}^N e_i^T(x, t) P^i B(f(z_i(x, t)) - f(w^*(x))) \\
&\leq \sum_{i=1}^N e_i^T(x, t) P^i B B^T P^i e_i(x, t) + \sum_{i=1}^N e_i^T(x, t) \Theta e_i(x, t) \\
&= e^T(x, t) (\hat{P} \hat{B} \hat{B}^T \hat{P} + \hat{\Theta}) e(x, t), \tag{24}
\end{aligned}$$

$$\begin{aligned}
&2e^T(x, t) \hat{P} (G^2 \otimes \Gamma_2) e(x, t - \tau(t)) \\
&\leq c_2 e^T(x, t) \hat{P} (G^2 \otimes \Gamma_2) Q^{-1} (G^2 \otimes \Gamma_2) \hat{P} e(x, t) \\
&\quad + \frac{1}{c_2} e^T(x, t - \tau(t)) Q e(x, t - \tau(t)). \tag{25}
\end{aligned}$$

It follows from (23)–(25) that

$$\begin{aligned}
\dot{V}(t) &\leq \int_{\Omega} e^T(x, t) \left[- \sum_{k=1}^q \frac{1}{l_k^2} (\hat{P} \hat{D} + \hat{D} \hat{P}) - \hat{P} \hat{A} - \hat{A} \hat{P} \right. \\
&\quad + \hat{\Theta} + \hat{P} \hat{B} \hat{B}^T \hat{P} + c_1 \hat{P} (G^1 \otimes \Gamma_1) + c_1 (G^1 \otimes \Gamma_1) \hat{P} \\
&\quad \left. + \frac{Q}{1 - \sigma} + c_2^2 \hat{P} (G^2 \otimes \Gamma_2) Q^{-1} (G^2 \otimes \Gamma_2) \hat{P} \right] e(x, t) dx \\
&\quad + 2 \int_{\Omega} e^T(x, t) \hat{P} u(x, t) dx.
\end{aligned}$$

Therefore,

$$\begin{aligned}
\dot{V}(t) - 2 \int_{\Omega} y^T(x, t) u(x, t) dx + \gamma \int_{\Omega} y^T(x, t) y(x, t) dx \\
\leq \int_{\Omega} e^T(x, t) \left[- \sum_{k=1}^q \frac{1}{l_k^2} (\hat{P} \hat{D} + \hat{D} \hat{P}) - \hat{P} \hat{A} - \hat{A} \hat{P} \right. \\
\quad + \hat{\Theta} + \hat{P} \hat{B} \hat{B}^T \hat{P} + c_1 \hat{P} (G^1 \otimes \Gamma_1) + c_1 (G^1 \otimes \Gamma_1) \hat{P} \\
\quad + \frac{Q}{1 - \sigma} + c_2^2 \hat{P} (G^2 \otimes \Gamma_2) Q^{-1} (G^2 \otimes \Gamma_2) \hat{P} + \gamma \hat{F}^T \hat{F} \left. \right] \\
\quad \times e(x, t) dx + 2 \int_{\Omega} e^T(x, t) (\hat{P} - \hat{F}^T + \gamma \hat{F}^T \hat{H}) u(x, t) dx \\
\quad + \int_{\Omega} u^T(x, t) (-\hat{H} - \hat{H}^T + \gamma \hat{H}^T \hat{H}) u(x, t) dx \\
= \int_{\Omega} \xi^T(x, t) \begin{pmatrix} \chi_1 & \chi_2 \\ \chi_1^T & \chi_2^T \end{pmatrix} \xi(x, t) dx,
\end{aligned}$$

where $\xi(x, t) = (e^T(x, t), u^T(x, t))^T$, $\chi_1 = \hat{P} - \hat{F}^T + \gamma \hat{F}^T \hat{H}$, $\chi_2 = -\hat{H} - \hat{H}^T + \gamma \hat{H}^T \hat{H}$. From (13), we can get

$$\dot{V}(t) + \gamma \int_{\Omega} y^T(x, t) y(x, t) dx \leq 2 \int_{\Omega} y^T(x, t) u(x, t) dx. \tag{26}$$

By integrating Eq. (26) with respect to t over the time period t_0 to t_p , we can obtain

$$\begin{aligned}
&2 \int_{t_0}^{t_p} \int_{\Omega} y^T(x, t) u(x, t) dx dt \\
&\geq V(t_p) - V(t_0) + \gamma \int_{t_0}^{t_p} \int_{\Omega} y^T(x, t) y(x, t) dx dt.
\end{aligned}$$

Namely,

$$\begin{aligned}
&\int_{t_0}^{t_p} \int_{\Omega} y^T(x, t) u(x, t) dx dt \\
&\geq \frac{1}{2} (V(t_p) - V(t_0)) + \frac{\gamma}{2} \int_{t_0}^{t_p} \int_{\Omega} y^T(x, t) y(x, t) dx dt
\end{aligned}$$

for any $t_p, t_0 \in \mathbb{R}^+$ and $t_p \geq t_0$. The proof is completed.

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